CVRP Mathematical Formulation — Component Summary

Component	Description	Formula/Notation
Graph	Directed/undirected, depot=0, customers 1..n	$G=(V,A)$, $V=\{0,\ldots,n\}$
Demand	Non-negative demand at each customer	$d_i \geq 0$, $i \in V \setminus \{0\}$
Capacity	Each vehicle carries at most Q units	$Q>0$
Cost/Distance	Travel cost on arc (i,j)	$c_{ij} \geq 0$
Decision var.	Binary: vehicle k uses arc (i,j)	$x_{ij}^k \in \{0,1\}$
Objective	Minimise total travel cost	$\min \sum_k \sum_{i,j} c_{ij} x_{ij}^k$
Capacity constr.	Load per route does not exceed Q	$\sum_{i \in S} d_i \leq Q \cdot \lvert routes(S) \rvert$
Flow constr.	Each customer visited exactly once	$\sum_k \sum_j x_{ij}^k = 1$, $\forall i$
Subtour elim.	Routes return to depot; no sub-cycles	Capacity cuts or MTZ constraints